CS 269 Communicating with Computers - Hackathon #1 (EvalGen)

Members: KK, Ashesh

The paper we chose to implement is “Who Validates the Validator?” by Shankar et. al. which proposed EvalGen, a pipeline to create criteria for alignment with user preferences. With the importance of transparent LLM-aided data labeling for both of our individual projects (knowledge graph intent labeling for KK, and automated workforce task delegation for Ashesh), we both decided that a robust means of generating criteria for LLM-guided classification will help improve our workflow.

Implementation

As such, we set out to implement EvalGen [source and artifacts]. First off, we fed Claude with the source paper and an unofficial implementation [source], asking Claude to re-create the paper that supports arbitrary LLM endpoints. Then, we fed the products to Codex, due to Claude reaching limits, to refine the product further to match the specifications listed in the paper, as well as the revibing paper. **Additional features** we added include: 1) **LLM selection**, should a different LLM be used for the initial generation and grading 2) **support for SVG** icon generation, as we aim to run the same EvalGen loop using non-textual input 3) run **export/import functionality** that allows past runs to be examined. The transcript for Claude and Codex can be found under the transcripts directory in the repository.

Revibing score

We computed **77.1% (54/70 points)**, for the current implementation at commit 82517fa. Using the author’s 35-test EvalGen rubric with the paper’s weights (full success = 2, partial = 1, failure = 0) we scored: 22x2 + 10x1 + 3x0 = 54 out of a maximum attainable of 70. The deductions are:

- **Missing controls:** table variable selector, keyword search, and scores-only toggle
- **Incomplete context:** input variables and response model badges are missing.
- **Interaction differences:** inferred criteria always default to LLM; manual entry retains previous suggestions; feedback appears for every grade; grading completion and Code/LLM presentation differ from the rubric.

Exports and visual grading are useful extensions, but earn no additional points under this rubric.

Experiences using EvalGen

We set out to use EvalGen for three tasks reproduces and extends the findings from the paper:

- **NER:** given a string of text (i.e. tweet), extract entities where there’s no best answer
- **Creative writing:** write a short description of a product for an advertisement. This measure alignment with the user’s preference on an open ended task
- **Logo design:** design a SVG logo. With newer LLMs being capable of code-based design, this tests the capabilities of EvalGen when used with modern tasks.

The actual prompts are listed in the repo. Each member provides their own text/product ideas. We used GPT-6.1-terra for both the seed LLM and the grader/criteria LLM.

Findings and Reactions

Our implementation of EvalGen can be used at https://cs269-evalgen.vercel.app. The results are in the repo. These are our reactions and findings:

KK: “I can see EvalGen simplifying both classification and prompting and I totally did not expect it working with SVG logo generation. I had it redesign the Google logo, only prioritized ones where the texts are colored, and the criteria of Integrated Colors popped up at 100% alignment”

Ashesh: “What surprised me was that for projects involving software engineering using coding agents, even though I had this project set up locally and reproduced, there was a sense of not having ownership over it. A lot of the cognitive load has shifted to validation in software engineering; if you make a mistake, it is easy to catch and see what you did wrong; however, for validation, mistakes are really difficult to catch.”



Figure 1: successful redesigns of popular logos (Google and Kodak) that won the criteria list.

For NER, we both find EvalGen to be similarly useful as the participants from the paper; criteria were inferred well from feedback with good alignment scores.

What surprised us was the dip in quality for creative writing, where we found that the criteria for an open ended task drift or fail to match the intention (i.e. presenting “must contain product name” as a criteria when all samples, regardless of user preference, contain product name). This led to a lower alignment score of about ~60-70-%.

For logo redesign, we found that EvalGen inferred criteria poorly; it felt like a lot of criteria that were inferred were not really the things that mattered. EvalGen achieved 0% overall alignment despite one individual criterion achieving 100%. We propose that a better aggregate score, perhaps one that weighs each criterion differently, will solve this alignment misscoring problem, as the one individual criterion is genuinely what ends up contributing most to user preference.

References

Adar, Eytan, et al. "Revibing Code from Papers: Reimplementing HCI Artifacts." *Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology*. 2026.

Shankar, Shreya, et al. "Who validates the validators? aligning llm-assisted evaluation of llm outputs with human preferences." *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 2024.